

Introduction_{*Honours*}

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1 Introduction

Climate change is increasing both the frequency and severity of wildfires by altering temperature, precipitation, and vegetation patterns [?]. Burnt area in EU countries has repeatedly exceeded historical averages in recent fire seasons [?]. Aerial firefighting (AFF) remains one of the primary tools available to suppress wildfires before they escalate, yet the coordination of aircraft fleets during an active fire is still largely dependent on the experience and judgement of human operators. Given the scale, time pressure, and safety stakes involved in a wildfire, there is a clear need for decision-support tools that can compute efficient aircraft routing and suppression strategies fast enough to be usable in live operations.

The Vehicle Routing Problem (VRP) and its variants have long been adapted to disaster and humanitarian relief settings, where capacity limits, multiple depots, and split deliveries are common features of the underlying logistics, as in the Split Delivery VRP [?] and drone-based relief distribution under payload and energy constraints [?]. chalabi2024affvrp adapt these ideas specifically to wild-fire suppression, introducing the Aerial FireFighting Vehicle Routing Problem (AFFVRP): a capacitated, multi-trip VRP with time windows and a hierarchical objective that first minimises time-to-last-drop and then total flight time, using a heterogeneous fleet of tankers and scoopers. liu2025hierarchical extend this direction with a hierarchical decision-making model for multi-aircraft *formations*, explicitly modelling the water-release process and tactical coordination behaviours that pure routing formulations do not capture. In both cases, however, the fire itself is represented as a static, pre-computed set of drop demands, with no feedback between the suppression actions taken and how the fire subsequently continues to spread.

Physically grounded fire spread models close part of this gap from the other direction. FARSITE [?] propagates a fire perimeter using Huygens' principle of wavelet expansion driven by the Rothermel surface fire spread equations, offering high fidelity with respect to fuel, terrain, and weather, but with no native mechanism for representing suppression actions. DEVS-FIRE [?] represents the burning area as a grid of discrete-event cells explicitly designed to accommodate suppression inputs, at the cost of the physical detail a continuous,

terrain-driven model provides. No existing formulation couples a routing and suppression optimisation directly and iteratively with a physically realistic fire spread simulator.

This work develops an optimisation framework for aerial firefighting that discretises the operational area into a finite grid of cells and a finite set of timesteps, allowing a routing and suppression model to be coupled directly with a fire spread simulator in a rolling-horizon loop. Each grid cell carries one of three states: unburned, burning, or extinguished. Fire propagation, aircraft flight dynamics, and water-drop suppression are all expressed over this shared discretisation. Aircraft are heterogeneous, following [?], water drops follow a hexagonal footprint approximated by a normal lateral distribution, and wind is treated stochastically to reflect operational uncertainty. The two objectives of minimising fire spread and minimising time to full extinguishment are retained from [?].

At the time of writing, a mixed-integer linear program solved with Gurobi implements this discretised formulation, including fire propagation, water-capacity dynamics for both aircraft types, and linear flight dynamics constraints. A rolling-horizon pipeline automates FARSITE (run inside FlamMap) to close the loop between suppression and continued fire spread. Several extensions are in progress: replacing the current probabilistic extinguishment rule with a deterministic, numerically-derived suppression criterion; allowing aircraft state to carry over continuously across rolling-horizon windows rather than resetting to the depot at each step; introducing a discrete-event simulation layer using salabim [?] to validate optimiser output before execution and reduce the computational load on the MILP; and migrating more detailed aircraft flight dynamics into this simulation layer.

The remainder of this paper is organised as follows. Section ?? presents the mathematical formulation of the discretised AFFVRP model. Section ?? [PLACEHOLDER — describe the FARSITE coupling / simulation methodology section]. Section ?? [PLACEHOLDER — describe the results/case study section]. Section ?? concludes and outlines directions for future work.